

1. **Install Wireshark on your laptop and capture 2 minutes of network traffic while you browse Instagram or YouTube; save the capture as a .pcap file and note the top 3 protocols you see.**

Steps

1. Install **Wireshark**.
2. Open Wireshark.
3. Select your **WiFi** network adapter.
4. Click **Start Capture**.
5. Browse **Instagram** or **YouTube** for about **2 minutes**.
6. Stop the capture.
7. Save the file as **network_capture.pcap**.

Protocol	Purpose
TCP	Reliable data transmission between devices.
TLSv1.3	Encrypts secure HTTPS traffic.
DNS	Resolves website names into IP addresses.

2. **Use the 'ping' command in Command Prompt or Terminal to test connectivity to www.spotify.com and www.zomato.com; record the response times and identify which site responds faster.**

```
C:\Users\hell0>ping www.spotify.com

Pinging atc.spotify.map.fastly.net [2a04:4e42:31::810] with 32 bytes of data:
Reply from 2a04:4e42:31::810: time=15ms
Reply from 2a04:4e42:31::810: time=15ms
Reply from 2a04:4e42:31::810: time=23ms
Reply from 2a04:4e42:31::810: time=17ms

Ping statistics for 2a04:4e42:31::810:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 15ms, Maximum = 23ms, Average = 17ms
```

```
C:\Users\hellO>ping www.zomato.com

Pinging e70929.dsca.akamaiedge.net [2600:140f:7200:49::17d4:67] with 32 bytes of data:
Reply from 2600:140f:7200:49::17d4:67: time=22ms
Reply from 2600:140f:7200:49::17d4:67: time=20ms
Reply from 2600:140f:7200:49::17d4:67: time=33ms
Reply from 2600:140f:7200:49::17d4:67: time=22ms

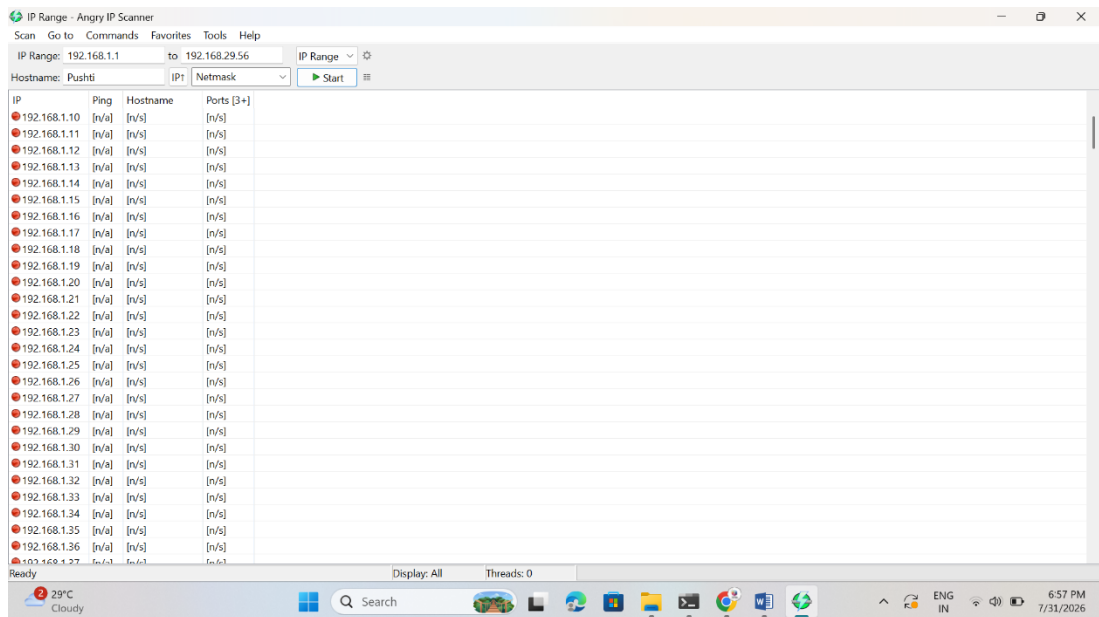
Ping statistics for 2600:140f:7200:49::17d4:67:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 20ms, Maximum = 33ms, Average = 24ms
```

3. **Open Task Manager (Windows) or Activity Monitor (Mac) and list all running processes related to Chrome, WhatsApp Desktop, or any browser-based app you use; note the process name and memory usage for each.**

Process Name	Memory Usage
Google Chrome	497
Chrome GPU Process	
Chrome Network Service	
WhatsApp Desktop	

4. **Download and run the free version of 'Angry IP Scanner' or 'Advanced IP Scanner' to scan your local WiFi network; list the IP and device name of at least 2 devices connected to your network

Hint: Make sure your firewall allows scanning, and only scan your own home/college network**



5. Use ChatGPT or Bing AI to suggest 3 hardware tools (like cable tester, crimping tool, or multimeter) used in network troubleshooting; for each, write one line describing when you would use it.

Tool	When to use
Cable Tester	To check whether an Ethernet cable is wired correctly and has no faults.
Crimping Tool	To attach or repair RJ-45 connectors on Ethernet cables.
Digital Multimeter	To measure voltage, continuity, and electrical faults in networking equipment or power supplies.